



FarDriver Controller APP Parameter Description

Complied by SiAECOSYS

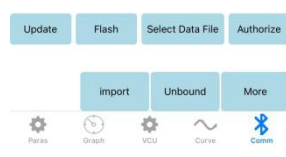
Non-profession do not operate!

1.Mobile host software interface:

When Bluetooth is not connected, the interface will be displayed as shown below, then "scan", search for the name "YQxxx", select this Bluetooth, and click "Connect".



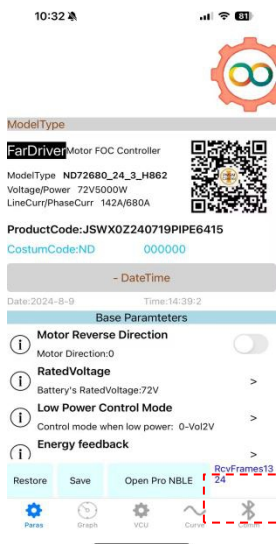
Bound:0031862 Received Frames:0



When the Bluetooth is connected, the interface is displayed as shown below, and then click "Bouding". The code of the corresponding binding controller can be seen in the binding list, which means that the binding is successful, and parameters can be modified normally.



After the binding is successful, the data is uploaded from the controller to the upper computer page, and the number of frames in the lower right corner will increase cumulatively, as shown in the figure:

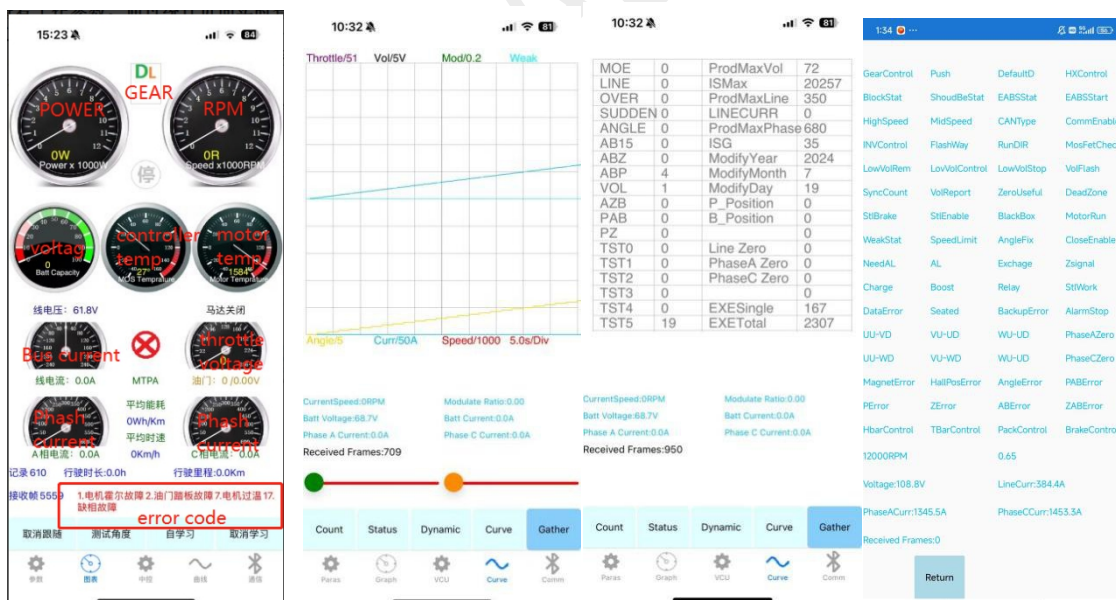


View 1 (Graph) contains a dashboard page for dynamic viewing of working parameters, real-time viewing of the dynamic characteristics of controller work through the statistics page, real-time viewing of controller work status through the status table, and clicking "Collection".

View 2 (curve) can view the operation data of 2048 groups of samples after the first acceleration, it is always stored in the controller if the controller kept turn on, such as 0.1 seconds to collect a set of data, and the controller can have 205 seconds of data records for analyzing the driving characteristics of the vehicle.

View 3 (count) is fault statistics during controller operation.

view 4 (status) shows the function enabled under the controller running status.



View 1

View 2

View 3

View 4

After binding the controller, click parameters, the display interface will display the model of the bound controller, voltage, power, DC current and phase current, product code, etc.



Basic parameters

Motor Reverse Direction: adjust the forward and reverse direction of the motor;

Rated voltage: the rated voltage value of the controller can be set;

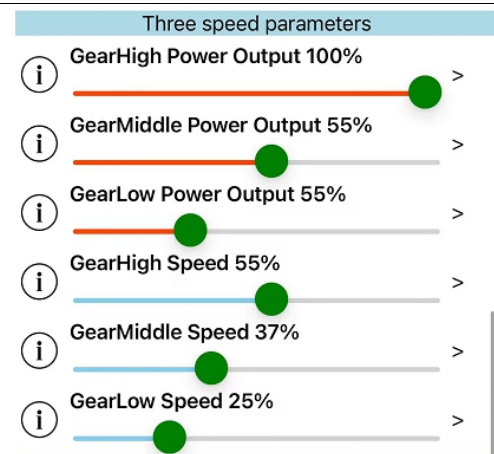
Loss control mode: the setting option to limit power output after the voltage reaches the "undervoltage value" in the set professional mode;

Energy feedback: set the mode of energy feedback;

Base Paramteters		
	Motor Reverse Direction	
	Motor Direction:0	
	RatedVoltage	>
	Battery's RatedVoltage:72V	
	Low Power Control Mode	>
	Control mode when low power: 0-Vol2V	
	Energy feedback	>
	Energy feedback Disabled	

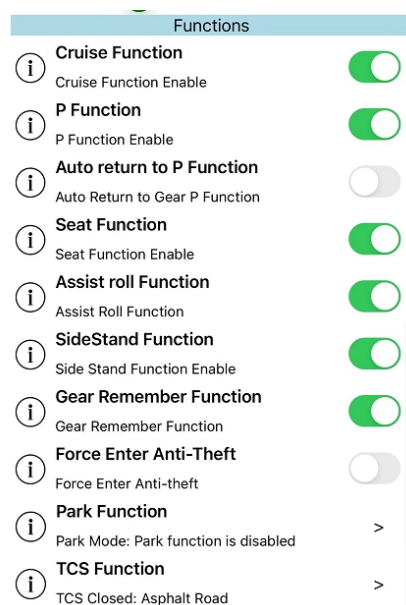
Three-gear speed parameters

Modify the current and speed limit ratio under the corresponding gear, which is consistent with the professional mode modification.



Function switch

The function is turned on and off. After turn on, the function port can be modified in professional mode.



2.Parameter

- Parameters	
AngleDetect: 0-120°Hall	TempSensor:5-KTY83-122
PhaseOffset:272	PolePairs:5
Motor Direction:0	RatedSpeed:5938RPM
RatedVoltage:72V	RatedPower:5000W
MaxSpeed:6700RPM	BackSpeed:230RPM
MaxLineCurr:142A	MaxPhaseCurr:680A
ThrottleResponse:0-Line	Throttle Acc Step:128
BoostLineCurr:150A	BoostPhaseCurr:680A
PhaseExchange: No Exchange	ECOAccCoeff: 8
Weak Character:0-Fast	WeakResponse:0
Throttle Dec Step:224	Release Throttle:0
Throttle Low:1.1V	Throttle High:3.95V

2.1 Basic Motor Parameters

2.1.1 Position sensor, the controller is divided to 4 kind of hardware according to the position sensor:

Email: info@siaecosys.com **Website:** www.siaecosys.com
Address:188 Kaifa Road east, Development Zone, Taizhou, Zhejiang, China

- 1) Hall version
- 2) Incremental encoder version
- 3) Differential encoder version
- 4) Absolute value encoder version
- 5) Spin encoder version.

No.	option	Controller hardware	Extension code	Edition
0	120° Hall	Hall controller		H,R,I
1	incremental encoder	Encoder controller		H,R,I
2	incremental encoder 4096	Encoder controller		H,R,I
3	incremental encoder 8192	Encoder controller		H,R,I
4	Differential encoder 2048	Differential encoder controller	Built-in J, external Q	H,R,I
5	Differential encoder 4096	Differential encoder controller	Built-in J, external Q	H,R,I
6	Differential encoder 8192	Differential encoder controller	Built-in J, external Q	H,R,I
7	Differential encoder 16384	Differential encoder controller	Built-in J, external Q	H,R,I
8-12				
13	Absolute value encoder 13	Absolute value encoder controller	Built-in H, external P	H,R,I
14	Absolute value encoder 14	Absolute value encoder controller	Built-in H, external P	H,R,I
15	Reservation			
16	60° Hall	Hall controller		H,R,I

2-15	2-15 Corresponding to the number of rotational poles	Spin encoder controller		X,U
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Need to click Save to modify the sensor. Hxx and later versions are optional. old versions are not optional.

Hardware version	Description
7	the oldest version, single function, relatively solidified
G	the olde version, single function, relatively solidified
A	the new version, support some parameter settings, single function, relatively solidified
B	the new version, support some parameter settings, single function, relatively solidified
H	the latest version, general, 30P, rich parameters
L	the latest version, general, Dual frequency , 30P, rich parameters
R	the latest version, national standard, Dual frequency , 6 Core +8 Core +16 Core socket, rich parameters
Q	the latest version, national standard, 6 Core +8 Core +16 Core socket, rich parameters
I	the latest version, isolated, 30P, rich parameters
J	the latest version, isolated, Dual frequency , 30P, rich parameters
X	the latest version, spin, 30P, rich parameters
U	the latest version, isolated spin, 30P, rich parameters
M	2024 New version, general, 30P, rich parameters
N	2024 New version, general, Dual frequency , 30P, rich parameters

There are many kinds of software programs, no longer one by one examples:

special code can be seen whether it is a special program:

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Address: 188 Kaifa Road east, Development Zone, Taizhou, Zhejiang, China

Code name	Implication
0~7	Hall vector starting speed
+8	8 Double PID
+16	Deep weak encoder program
+32	Brake detection
+64	Old controller
+128	No Bluetooth RS485 Communication program
+256	Series number 8 Special procedures
+512	Series number 12 Special procedures
+1024	Series number 16 Special procedures
+2048	Isolated version of the program
+4096	120M High-speed version of the program

In addition, there are various specific programs specially required by users, which are not reflected in the code name. Controller function code:

Code name	Implication
'H'	Deluxe version
'S'	Track version
'X'	The 'X' charging protection signal is reversed, while the non-'X' charging protection is effective at a low level.
'V'	National standard car, default 36V, other national standards by default 48V

2.1.2 Temperature sensor:

option	Hxx version	Old version
Not have	√	√
PTC-1000	√	√
NTC-230K	√	√
KTY84-130	√	Fixed program KTY84-130/KTY83-122
Hypothetical	√	√
KTY83-122	√	×
NTC-10K	√	×
NTC-100K	√	×

2.1.3 Phase shift:

For the key characteristics of the motor angle position, the general motor factory will indicate the angle position. Wheel hub motors on the market most cities 30°, 210° and There are three types of 90°, but some motors are special. Note that the motor factory standard method is different from the long-drive logo, if This perspective is not clear, and this value can be found through self-study. Generally, the two phase shift errors do not exceed 2° proves that there is no problem with the operation method, and there is no problem with phase shift.

2.1.4 number of pole-pairs

Hall's motor default 4 does not need to be changed. Encoder motor pole number setting must be accurate otherwise can not turn. Optional values = 3,4,5,6,7,8,10,12,14,16 ~30, encoder: 3-8 pairs of poles to display the actual speed, more than 10 pairs of poles to display the speed of 4 pairs. Modify the number of poles, save the key points only valid.

2.1.5 Motor direction:

Specify the direction of the motor when moving forward. 0: Motor on the right, 1: Motor on the left. It is only effective after the reset is saved.

2.1.6 Rated speed: the speed of the motor at rated voltage, referred to as rated speed, is often called in the electric motorcycle industry. It is a fixed speed.

This fixed speed determines the highest motor speed. General ordinary controller, at the rated voltage state, can Drive electricity The maximum speed of the machine is near the fixed speed. The controller will recognize the current power during self-learning. The rated speed under pressure.

2.1.7 Rated voltage:

The maximum number of series of batteries for FarDriver controllers with different voltages is as follows:

	Lead-acid battery	Ternary lithium battery	Lithium iron phosphate battery
48V	4 Strings	13-14 Strings	16 Strings
60V	5 Strings	17 Strings	20 Strings
72V	6 Strings	21 Strings	24 Strings
75V	6 Strings	22 Strings	25 Strings
84V	7 Strings	24 Strings	28 Strings
96V	8 Strings	28 Strings	32 Strings
108V	9 Strings	32 Strings	35-36 Strings

Factory setting 72 Series = 72V, 75 Series = 75V, 84 Series = 84V, 96 Series = 96V, 108 Series = 108V

Note that the rated voltage affects the power display. The setting cannot be higher than the factory voltage. After setting the parameters, click Save again. It is effective after the position.

2.1.8 Rated power: Please set the rated power of the motor according to the actual situation of the motor. The AB version is not allowed to be changed..

2.1.9 Maximum rotational speed: The maximum speed of the motor is limited.

1) In the electric vehicle market, the maximum speed is generally not limited, but the maximum speed is limited by the following current limiting parameters. After the speed exceeds the fixed speed, Automatically enter the weak magnetic state. The more the speed exceeds the fixed speed, the greater the weak magnetic depth.

2) Weak magnetic depth: $(\text{Maximum speed} - \text{consted speed}) / \text{Fixed speed} * 100\%$. Generally, the weak magnetic depth of the hub motor can be reached. 50%.

Some hub motors have weak magnetic depth. More than 100%. Therefore, we stipulate that the weak magnetic depth of the tape motor should not exceed 50%, The weak magnetic depth of the embedded motor does not exceed 150%.

2.1.10 Maximum phase current:

The maximum phase current of the working motor. It determines the maximum torque of the motor output at the rated speed.

The maximum phase current is limited by the controller hardware, and the set value is not allowed to exceed the factory settings. Otherwise, it will guide. To control the probability of the machine burning is greatly increased.

Different types of motors will have different output torques under the same maximum phase current setting value. Turn round moment version The output torque of the motor is large, the output of the balanced version is slightly smaller, and the output of the speed version is the smallest. The output torque of the motor with low fixed speed is large. The output torque of the motor with high fixed speed is small.

2.1.11 Maximum line current:

The maximum bus current of the operating battery of the controller. The maximum power value of the motor output is determined. Maximum input power of the controller = battery voltage * maximum line current.

This current is limited to the maximum line current of the customer. This value determines the maximum output power, thus the highest.

2.1.12 Retrest speed:

The maximum speed of the backward gear.

2.1.13 Switching phase line:

Tacitly approve 0, if the blue-green line is exchanged, it is 1. Note that if this parameter is not correct, the motor will not rotate. It is valid after reset. Hxx The version self-learning will be automatically modified. This parameter.

2.1.14 Weak magnetic characteristics:

Generally fast, the high-speed jitter is changed to medium. Generally, it is not used slowly, and it is easy to cross the current.

2.1.15 Weak magnetic response:

0~6, no means weak magnetism. Default weak magnetic response 0

Expansion speed: Push the motor speed to a higher speed than the fixed speed, which is called expansion speed.

Speed expansion method 1: Increase the working voltage, the higher the voltage, the higher the motor speed.

Speed expansion method 2: Do not increase the working voltage, but increase the motor speed through weak magnetism. Without changing the battery voltage, the motor speed can be increased directly by controlling the current limiting parameters.

2.2 Acceleration and deceleration characteristics

2.2.1 Accelerated sensitivity:

Acceleration speed, 8~224, The larger the number, the faster the accelerator response.

Electric vehicles are generally accelerator pedals, while electric motorcycles are accelerator handlebars or central control.

Electric vehicles should react moderately to the throttle, while the requirements of electric motorcycles are different. Some customers require light, slow, steady, Some customers require responsiveness and immediate response.

Acceleration sensitivity refers to the speed of the throttle response. This parameter is in 16~224 Between. The larger the number, the faster the accelerator. The more sensitive it is.

16 It's already very slow. It's usually set up on electric vehicles. 32 It is suitable for left and right, and rarely exceeds 64.

For electric motorcycles, in addition to setting in 32 In addition, Many users prefer to respond quickly, so it is set in 64, 128. The track race is even set in 224.

2.2.2 Deceleration sensitivity:

Deceleration: 16~224, the larger the number, the shorter the throttle lag.

2.2.3 Back to the accelerator:

Tacitly approve 0

2.2.4 Throttle response:

For different user preferences, there are three configurations: linear, sports, and economic.

2.3 Throttle threshold

The handlebars on the market are uneven, and the voltage values of different handlebars or accelerator pedals will be different.

	Idle voltage	Full voltage
Electric motorcycle Throttle	0.8V-0.9V	4.1-4.3V
Central control Throttle	0.8V-0.9V	4.5-4.95V

12V Throttle pedal	0.0V-0.2V	4.6-4.8V
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2.3.1 Low threshold:

We set the low throttle threshold according to the idle voltage. Considering the fluctuation of the handle voltage, it is generally necessary to set a low throttle threshold. Higher than idle voltage 0.2-0.3V to ensure that when stopping the motor is working in an idle state.

For example, the low throttle threshold of the electric motorcycle handle will be set to 1.1V, and 12V The low throttle threshold of the accelerator pedal will be set to 0.5V.

2.3.2 High threshold:

We set the high throttle threshold based on the full handle voltage. In order for the controller to output the full power rate in the full bar state, we need to make the set value lower than the full voltage. But be careful not to set it too low. In order to automatically detect whether the electric throttle is damaged, we set a value 0.6V higher than the high throttle threshold as the alarm limit, once it is exceeded, it is considered that the handle is damaged, and the controller immediately stops the power output to avoid the vehicle racing and avoid causing the safety accident of the car.

Therefore, when we set a high throttle threshold, for example, when the electric motorcycle turns the handle full 4.1-4.3V, we will set 3.9V as the high throttle threshold. For the high throttle threshold of the 12V gas pedal, we will set it at 4.3V. 742 and above version added the throttle self-learning function, self-learning to the end, the controller will automatically identify the maximum voltage of the throttle signal of the crank/pedal, and generate the throttle high threshold according to this voltage.

3. Proportional coefficient of speed limit

- Ratios in Speed	
500RPM 100%	1000RPM 100%
1500RPM 100%	2000RPM 98%
2500RPM 96%	3000RPM 94%
3500RPM 92%	4000RPM 90%
4500RPM 88%	5000RPM 85%
5500RPM 83%	6000RPM 80%
6500RPM 50%	7000RPM 15%
7500RPM 0%	8000RPM 0%
8500RPM 0%	9000RPM 0%
LD:550	LQ:329
FAIF:513	LimitSpeed:7000RPM
- Ratios in Gear	
LowSpeedLineRatio: 31%	MidSpeedLineRatio: 55%
LowSpeedPhaseRatio: 55%	MidSpeedPhaseRatio: 66%
LowSpeed:3000RPM	MiddleSpeed:4500RPM

3.1 Motor current limiting protection coefficient:

In the flow limit 500RPM,1000RPM,.....8500RPM,9000RPM The conversion.

Note that the current limiting parameters have new meanings, in LD=100 At that time, the coefficient at each speed is Laserdisc Value = coefficient*9;

These rotational speeds are based on the number of poles calibrated in the parameters. The corresponding parameter is also the speed. The parameters below. And for the actual polar logarithm is greater than or equal to 16 The motor needs to be converted.

Usually, the polar logarithm of the hub motor is 16, 20, 24, 28, 30 Right. Usually, the middle motor is 3, 4, 5, 7, 8, 14 Right.

The polar logarithm in the parameter = 4 In this case, the motor speed = the speed on the upper computer * 4 / the actual logarithm of the motor. For example, the polar logarithm of the hub motor is 16

Like that

500RPM Corresponding 16 The actual speed of the polar hub motor is 125RPM.

1000RPM Corresponding 16 The actual speed of the polar hub motor is 250RPM. 4000RPM Corresponding 16 The actual speed of the polar hub motor is 1000RPM. 5000RPM Corresponding 16 The actual speed of the polar hub motor is 1250RPM. 5500RPM Corresponding 16 The actual speed of the polar hub motor is 1375RPM. 6000RPM Corresponding 16 The actual speed of the polar hub motor is 1500RPM. 6500RPM Corresponding 16 The actual speed of the polar hub motor is 1625RPM. 8000RPM Corresponding 16 The actual speed of the polar hub motor is 2000RPM

Weak magnetic limit: gradually increase the current limit parameters

The current limit value is set from the safety value, gradually increase the speed, and be sure to ensure that the weak magnetic field cannot be excessive. Once the idling speed is found to be unstable or even out of the line MOE or OVER protection, it indicates that the speed is too high, the weak magnetic degree is too high, and the parameters should be changed back.

The current limit value we set should be considered according to the actual demand. For a motor with a constant speed of 1000RPM, the weak magnetic depth is considered to be 50%. The maximum speed is also considered at 1500RPM, and it is hoped that the motor will not work above 1625RPM. Therefore, the current limit value is set at 30% at 6000RPM, and the speed of 6500RPM and above is set at less than 5%. This ensures that the motor is 50% weaker when idling. It will not cause motor jitter or even burn control due to weak magnetic depth.

Many motors, weak magnetic depth can reach 100%, 1000RPM motor can work at 2000RPM high speed. For this kind of motor, in order to play a higher performance, you can continue to expand the current limiting coefficient, the current limiting parameters within 8000RPM can be set above the normal value of 70, 8500 is set at 30, 9000RPM is set below 5.

3.2 Speed control level:

Speed control level: default 4 Level: BOOST Gear, high speed file, Medium speed, low speed.

1) BOOST File: Mobile phone APP/Displayed on the computer bst. At BOOST The function is effective. BOOST Time by 4.3 Parameter control, BOOST Limit the maximum line current, maximum phase current and current limiting coefficient of customers who leave the factory. Condition Next to work.

2) High-speed transmission: mobile phone APP/display on the computer D. Tall. The speed gear is subject to the maximum line current, the maximum phase current and current limiting coefficient. And work under the maximum speed limit.

3) Medium speed: mobile phone APP/display on the computer DM. Medium speed .The gear is proportional to the medium-speed line current and the medium-speed phase current ratio. And limited by medium speed.

4) Low-speed transmission: mobile phone APP/display on the computer DL. Low-speed gear with low-speed line current ratio and low-speed phase current ratio And limited by low speed.

3.2.1 LDVOL

200-900, default 900,

The matching of idling noise in the high-speed section and the balance between the highest power and power saving.

Special 100: Through each speed Laserdisc Restrictions, The biggest 100 Match Laserdisc Value 900.

3.2.2 LQVOL

1) Intermediate throttle voltage (mV, VQH+4 The middle throttle is enabled, 0~4000).

2) Speed limit current, (0~4000, unit 1/4A)

3.2.3 FAIF:

0~513	0	Starting resonance characteristic matching option 1 Pay attention to when the motor performance of the vehicle is good, VQH+64=Six times the speed Hall detection It can further reduce resonance noise.
	16~63	Starting resonance characteristic matching option 2
	64~256	General vehicle motor Hall characteristics
	384	Jumping car AB15 Choose when it is easy to report an error. 384/512.

	512	
	513	CN The absolute value of the controller is ignored in the case of the encoder AB15 Alarm.
		Other controllers: easy AB15 Use it when reporting errors.
+1024	Secret code	1 Indicates that the fixed Bluetooth password is related to the product number, 0 The default user sets the password
+2048	Retain	
+4096	Retain	
+8192	TCS Prohibit	1 Indicates prohibition TCS Function, 0 Turn on by default TCS Function
+16384	Retain	
+32768	Retain	

3.2.4 Speed limit: the speed used for the speed limit of the Morse code.

3.3 Three-speed speed-limiting ratio

3.3.1 Low speed parameters

Low-speed line current ratio, low-speed phase current ratio, low-speed rotational speed

3.3.2 Medium-speed parameters

Medium-speed line current ratio, Medium-speed phase current ratio, Medium speed

4 . Energy feedback

- Energy Regenerate

StopBackCurr:2A	MaxBackCurr:4A
Batt RatedCapacity:20AH	
FreeThrottle 0	
500RPM 0%	1000RPM 0%
1500RPM 0%	2000RPM 0%
2500RPM 0%	3000RPM 0%
3500RPM 0%	4000RPM 0%
4500RPM 0%	5000RPM 0%
5500RPM 0%	6000RPM 0%
6500RPM 0%	7000RPM 0%
7500RPM 0%	8000RPM 0%
8500RPM 0%	9000RPM 0%

For the return throttle brake function, during the ride, Enter after returning to the accelerator. Enter the electronic brake state.

For the electronic brake function, when braking, the whole car gives a brake signal to the control. The brake signal is detected by the controller. After the number, it will enter the electronic brake state.

Note that when you want to use the electronic brake function, you must select the electronic brake or return throttle brake in the following item to enable this. Function. And set the reflux current. Note that the maximum reflux is generally more than stopping when setting parameters. Stop the reflux. 25%~50%.

4.1 Brake current limit:

4.1.1 Brake feedback current:

The brake current of the electronic brake. Tacitly approve 2A, when the electronic braking force is required, change it as needed. 5A-20A, for The large-capacity battery of the four-wheeled vehicle has a larger back-charging current and can be set to 20A-60A.

4.1.2 Maximum reflux:

Peak brake current of electronic brake: default 4A, when you need strong braking, it can be set to 10A-40A, for four- wheelers It can be considered 40A-80A.

4.1.3 Battery capacity: It can be set according to the actual battery capacity.

4.2 Negative current coefficient:

Tacitly approve 0, the faster the return speed of the accelerator, the smaller the braking force. Slow speed and light braking force. 1: The brake force is the strongest when returning to the accelerator.

Above this speed, turn back half of the speed without adding a uniform speed. The speed does not slow down. Take for example 4000 Express 0- 4000 The higher the rotation speed, the closer to half of the throttle value, 4000 The above intermediate throttle value is uniform and does not accelerate.

Deceleration state. Speed up above the middle value, below the middle value, slow down, and the more you turn back. The more powerful the brakes are.

Control the proportional coefficient of the reverse charging current, in 500rpm, 1000rpm, ... ,9000rpm Control at each speed. The maximum coefficient is 0, the minimum value is -100. The closer it is to -100 Negative electricity .The more flow.

Ordinary two-wheeler, for 400A The left and right controllers have a negative current coefficient of -10%~-30%. Other controller roots According to the adjustment of the situation, for driving safety, you can start debugging from -10%. If you think it is not enough, change it to -15%. - 20%, don't set it to -50%~-100 at once.%, such an operation is easy to bring the risk of sudden braking.

The old version of the controller does not have a negative current system.

5.Function

- Functions	
Boost Pin:13-Invalid	Cruise Pin: 8-PIN17
SideStand Pin:13-Invalid	Pause Pin: 0-NC
Forward Pin:13-Invalid	Backward Pin: 4-PIN8
Highspeed Pin: 2-PIN3	LowSpeed Pin: 1-PIN2
Charge Pin:13-Invalid	Anti-theft Pin: 6-PIN14
Seat Pin:13-Invalid	SpeedLimit Pin:13-Invalid
Switch Voltage Pin:13-Invalid	Repair Pin:13-Invalid
BoostTime: 10s	BoostRelease: 35s
HighLowSpeed: 0-HighOnly	Puse RE:0-Invalid
EmptyRun: 0	SlowDown: 3
Gear:1-Default D	
Brake:0-StopWhenGround	PC13:0-NomalResponse
Park:2-Disabled	Follow:1-Disabled

5.1.1 BOOST: Choice BOOST Function button input foot .If the selection is invalid, this function will not be enabled.

5.1.2 Cruise: Select the cruise function button to enter the foot. If the selection is invalid, this function will not be enabled.

5.1.3 Side support: Choose support at the same time function button input pin, select invalid, this function will not be enabled.

5.1.4 Parking lot File: Select Parking lot File function button input foot, select **Normally closed** Will not enable this function

5.1.5 Forward: Select the forward function line input foot. If the selection is invalid, this function will not be enabled.

5.1.6 Back: Select the back file function line input pin. If the selection is invalid, this function will not be enabled.

5.1.7 High-speed: Select the high-speed function line/button input pin, and the selection is invalid. This function is not enabled.

5.1.8 Low speed: Select the low speed function line input foot. If the selection is invalid, this function will not be enabled.

5.1.9 Charging: Select the charging protection function cable input foot. If the selection is invalid, this function will not be enabled.

5.1.10 Anti-theft: Select the anti-theft function function line input pin. If the selection is invalid, this function will not be enabled.

5.1.11 Seat bucket: Select the seat bucket function line to enter the foot, and when sitting, remove Parking lot Drive normally after the gear, no you can't drive.

If the implementation function is enabled, start the cart assistance when no one is sitting. Select the normal closure according to VCU Instruction recognition sitting Bucket signal and implementation function. Selecting invalid will not enable this function.

5.1.12 Speed limit: Select the speed limit function line input foot, and the invalid selection will not start. Use this function

5.1.13 Voltage switching: select the rated voltage switching function line input enter the foot, select invalid, this function will not be enabled.

Voltage switching foot = PIN8 At that time, the voltage foot of the national standard speed limit is switched, and the speed foot is special. Definition

5.1.14 One-click repair: Select one-click repair function line input foot, Selecting invalid will not enable this function. Pay attention to the selection switch

When parking, one-click to repair the foot as the switch for this function.

Pay attention to Parking lot The file may choose to be normally closed, and other feet generally do not choose to be normally closed, otherwise it will affect the function. Use. Here is the optional pin definition description

option	6/12 tube NS series	Common Hall ports for older controllers	More than 18 GB 485 controller	YJCAN	YJBMQ CAN	Old coders connector	Remarks
Normally closed	Normally closed	Normally closed	Normally closed	Normally closed	Normally closed	Normally closed	

PIN2	2 Foot	-	-	-	-	-	PB9
PIN3	3 Foot	7 Foot	24 Foot	3 Foot	3 Foot	3 Foot	Personal computer14
PIN5	5 Foot	5 Foot	3 Foot	-	-	5	CAN The version cannot Use PA12
PIN8	8 Foot	8 Foot	14 Foot	8 Foot	8 Foot	-	Personal computer15
PIN9	9 Foot	-	-	-	-	-	PB5
PIN14	14 Foot	17 Foot	18 Foot	2 Foot	2 Foot	-	Personal computer13
PIN15	15 Foot	4 Foot	2 Foot	-	-	4	CAN The version cannot Use PA11
PIN17	17 Foot	30 Foot	-	17 Foot	-	-	The encoder version is not It can be used PA15
PIN18	18 Foot	-	-	14 Foot	14 Foot	-	PD0
PINr24	24 Foot	25 Foot	15 Foot	24 Foot	24 Foot	-	PA0
PD1							PD1
PB4							PB4
Be invalid	Be invalid	Be invalid	Be invalid	Be invalid	Be invalid	Be invalid	

5.2 Special function

5.2.1 High and low speed:

- 1) High-speed only: only high-speed transmission is required
- 2) Plus and subtract gears: button plus and subtract gears
- 3) Point movement high and low speed: point movement 2 Speed: high speed + low speed
- 4) Click High Speed: Click 2 Speed: low speed + medium speed, (H58 Version)
- 5) Point movement three speed low: point movement 3 Speed, default low-speed gear
- 6) Click to move at three speeds: click to move 3 Speed, default medium speed
- 7) Click three-speed high: click 3 Speed, default high-speed gear

- 8) Point four-speed low: click 4 Speed, default low-speed gear
- 9) Move at four speeds. 2: Click 4 Speed, default 2 Speed gear
- 10) Move at four speeds. 3: Click to move 4 Speed, default 3 Speed gear
- 11) Point movement four-speed high: point movement 4 Speed, default high-speed gear
- 12) Dial three-speed: dial 3 Fast
- 13) Serial port gear: serial port control gear, default low-speed gear The serial port instrument of the moped.XM,
- 14) CAN Gear:CAN Control gear, default low-speed gear
- 15) Be invalid 5.2.2 Long press back:

When it is valid, you must press the back button for a long time to switch to backward.Tacitly approve Invalid, switch back according to the back line.

5.2.3 Gear:

The gear	Explain
0-default neutral	Start by default N File,FW Grounding = moving forward,RE Grounding = backward
1- Advance by default	Start the default forward file, RE Grounding = backward, you can have Parking lot File (same as N)
2-Reverse gap	FW Ground = neutral, otherwise RE Grounding = backward,RE Suspension = forward
3-Default click low	H Series:FW Enter the anti-theft alarm lock when it is suspended. Grounded driving is allowed. Other series: default advance low speed
4- Default click-in-motion	Other series:Move at medium speed by default and click.
5- Default point movement height	Other series:Default advanced high-speed, click

5.2.4 Brakes:

Brake function: You can drive when you don't pinch the brake, and disconnect the accelerator when you pinch the brake.

Floating driving: high brake line connection when pinching the brake 12V Or battery voltage. Or the low brake foot is connected. Battery floor. Pay attention to the brake The line signal is not isolated. When you don't brake, 2 All lines should be floating.

Floating power failure: high and low brakes when pinching the brakes 2 The thread is floating in the air. When the brake is not pinched, the high brake line is connected to 12V. Or the battery Voltage, or low brake foot to turn on the battery, pay attention to the brake line signal is Not isolated.

P+ floating driving: In addition to the floating driving function, the brake is lifted at the same time. Parking lot File status. P+ floating power off: In addition to the floating power off function, the brake is removed at the same time. Parking lot File status.

Be invalid: Floating driving by default.

5.2.5 PC13:

Parameters of the old controller: floating driving, floating power failure, floating cruise, grounded cruise, invalid. H Series controller parameters: normal response and track response.

5.2.6 Station on the slope:

1) Hang the gear to prevent slipping on the slope: When moving forward or backward, release the accelerator and it will stop on the slope. The gap does not stay on the slope, and the steep slope falls slowly. The function is invalid.

(H937 Later version, you can move forward and backward on the slope, steep slow down the slope and adjust the retreat distance 1 Long distance, 7 Distance Short, slow-down coefficient 2~15 Adjust the oscillation damping. Approve in order to choose the brake not to stop the car, the parking will be cancelled after changing gears. It is very smooth. If adjusted properly, the back slip distance can be shortened to the length of a cigarette., the most power-saving way to stop the car)

2) Parking lot Gear switch on the slope: in Parking lot In the file state, it is stationed on the slope, and in other states, it is set to start according to the slow descent parameters of the steep slope. Use the steep slope slow-down function. (1) H935 Version and above, (2) One-click to repair the foot setting is normally closed. (3) Then adjust Slow descent of steep slopes 0~7, [0 No current consumption, 7 The maximum current consumption is the maximum parking intensity. Pay attention to the non-station. Adjustment under the car], (4) Change the current anti-theft to invalid, (5) Pinch the brakes 2 It will take a second to enter. 6) The gas door is removed from parking. (7) Open The characteristic of closing the parking is that after pinching the brake and then loosening the brake, the parking car does not move at all, and the effect is very ideal.

3) Invalid: The standing function and the slow descent function of steep slope are invalid.

4) Slow landing parking:

5.2.7 Follow:

Follow: The motor enables a certain idle speed Invalid: shielding follow-up and electronic brake

Electronic brake: Start the electronic brake when pinching the brake. Return to the accelerator brake: start the electronic brake when returning to the accelerator

5.2.8 Slow descent of steep slopes:

None: Do not enable the steep slope to slow down.

1~7: The smaller the number, the slower the slow-down reaction, and the larger the number, the faster the slow-down reaction.

Parking effect: steep slope slow descent number small retreat will be more but stable, number the words are big and retreat less, but they rush forward too much.

5.3 BOOST

5.3.1 BOOST Duration:

BOOST The duration after startup is default 45 Seconds, the maximum 131 Seconds;

5.3.2 BOOST Interval:

BOOST After it's over, start again. BOOST The time required is required, by default 90 Seconds, the maximum 131 Seconds;

5.3.3 Idoi:

Default 0

6.Appearance

- Display	
Speed Pulses:4	SpdPulseNum:40459
SpeedoMeter:0-Pulse	AnalogSpeedCoeff:10000RP
CAN: CAN60	CAN Detect: 150ms
CAN Baud:0~250K	TorqueCoeff:7420
Step:0~0.5ms	Stop:1~55ms
SpecialFrame: 0	PULSE: 0
SQH: 0	DATA0: 8
DATA1: 97	SEC0: 0
SEC1: 0	SEC2: 0
SEC3: 0	SEC4: 0
SEC5: 0	SEC6: 0
SEC7: 0	PPosition: 1
BCPosition: 3	HbarPosition: 0
FDPosition: 8	CurrentCoeff: 64
ByteOption:0	WheelWidth: 120
WheelRatio: 70	WheelR: 12
GearRatio: 4	

6.1 Instrumentation method

The controller has 3 A signal output foot, among which

12 Tube controller and NS Series controller: 13 Foot RXD, 18 Foot ALARM/SPD, 9 Foot Spa

Old version of the controller: 3 Foot RXD, 9 Foot SPD, 10 Foot SPA, H The extension code of the series controller will be marked with characters. A~G

6.1.1 Speed pulse:

This value 1-31, affecting the pulse speed output and the first-line speed display. The larger the number, the higher the display speed of the table.

6.1.2 Speed pulse base:

The calibration base of the speed pulse meter. Changing this value only affects the speed. The speed display of the pulse meter. Wheel hub default

40459, Middle default 26043

6.1.3 Speedometer method:

Pulse/simulation/isolation pulse

6.1.4 Analog speedometer:

The voltage of the phase line instrument indicates the coefficient adopted by the instrument of speed, and adjust this coefficient. The speed of the display can be changed.

The maximum speed of remote control = $500 * \text{analog speedometer} / 1024(\text{RPM})$;

6.1.5 CAN:

Instruction number, Hxx Version default 60, The previous version has different instruction numbers according to different protocols. CAN=59: Support driver less system, special CAN Configuration parameters

CAN=48: (H80 Version) Switch from serial port to CAN Debugging of analyzer.

6.1.6 CAN Detection delay:

Tacitly approve 150 Master of Science Individual customer requirements 1900 Master of Science

6.1.7 CAN baud rate

0-250K 1-500K 2-1M

6.2 First-line parameters

The first-line display should pay attention to the pin not be wrong: 12 The following general choices 18 The foot line is connected, and the configuration of individual drivers is shared. 13 The feet are on the line. NS

Series 18 The controller above is the same as 12 Tube. 18 The old version of the controller above is general 9 Foot one The line is open.

6.2.1 step length:

Most line scores are 0.5 or 0.9. Default 0.5, optional 0.9,1.2, 1.9.

6.2.2 Stop interval cycle:

Most first-line services support 55ms, default 55ms, optional 24 ms, 144 ms, 216ms;

6.2.3 PULSE :

0 by default, non-0 by custom.

6.2.4 SQH:

0 by default, non-0 by custom.

6.2.5 Special frames:

The following 0-255 define special frame types.

	Special frame	Explain
1	0~1	Non-front-line communication 0 : Fast Linear measure Arteries and veins Thoroughfare , (Obvious Show Transfer Node : Fast Linear measure Arteries and veins Thoroughfare Base Count Jump over Big Obvious Show Fast Linear measure Jump over Slow 500~65530,V39 And the previous version is 5000~65530) 1:READY Lamp: (READY The state is high output, otherwise the output is low) 2: Fan control: (the temperature is lower than 40° is the output high, higher than 40° is low output) 3: Special serial port instructions BF_ZHULI 4: Special serial port instructions ZHULI Assisted pulse detection (PIN3) 5: Special serial port instructions KM5 6: Special serial port instructions UK1 7: Special serial port instructions Step length, interval time,PULSE=0,SQH=0,DATA0-DATA1,SEC0-SEC7 Be invalid

2	16~31	General Front Line 2 Different instruments,DATA6/DATA9/DATA10 The content to be sent is different. Please refer to the Communications Association. Discuss the content of the first choice: DATA6: Byte option = 3 Time output 0, otherwise the output current value. Refer to the table below DATA9 Options: Refer to the following DATA9/DATA10 List of options DATA10 Options: Refer to the following DATA9/DATA10 List of options Special frame = 16+ DATA9 Options + DATA10 Option, general first-line pass = 21 <table><tr><td>Parameter</td><td colspan="6">List 6 Group parameters, different first-line instruments, their parameters are not similarly, the first set of</td><td></td><td></td></tr></table>								Parameter	List 6 Group parameters, different first-line instruments, their parameters are not similarly, the first set of							
Parameter	List 6 Group parameters, different first-line instruments, their parameters are not similarly, the first set of																	
			parameters is commonly used.							It's normal.								
		Parameter group	1	2	3	4	5	6	7									
		PULSE	0	0	0	0	10	0	0									
		SQH	0	0	0	0	1	0	0									
		DATA0	(8) 0x08	(89) 0x59	(24) 0x18	(3) 0x03	(16) 0x10	(81) 0x51	(16) 0x10									
		DATA1	(97) 0x61	(66) 0x42	(2) 0x02	(1) 0x01	(18) 0x12	(0) 0x00	(149) 0x95									
		Step length suggestions 0.9Master of Science Suggestions on the duration of the interval 144Master of Science Science, there are some needs. 216Master of Science SEC0~SEC7 Default full 0																
3	32~40	No line, built-in Bluetooth Special frame: 32-TBIT, 33-XZ_CONTROL, 34-XMZSBXX, 35-XM3SPEED, 36-M2S, 37-CN Step length, interval time,PULSE=0,SQH=0,DATA0-DATA1,SEC0-SEC7 Be invalid																

4

48~223

Encryption first-line, internal SEC

Different instruments,DATA6/DATA9/DATATen The content to be sent is different
Please refer to the Communications Association. Discuss the content of the first
choice:

Base number = 48 64 80 96 112 128 144 160 176 192 208

Cardinal number	DATA0	SEC0	...
48	0X08	0X6B	...
64	0X07	0X9C	...
80	0X30	0X73	...
96	0X27	0X0E	...
112	0X10	0XBA	...
128	0X2B	0X2C	...
144	0X5	0XB2	...
160	0X5	0X2B	...
176	0X25	0XEA	...
192	0X0A	0X2C	...
208	0X1F	0X9E	...

DATA6: Byte option = 3 Time output 0, otherwise the output current value.

DATA9 Options: Refer to the following DATA9/DATA10 List of options.

DATA10 Options: Refer to the following DATA9/DATA10 List of options

Special frame = base +DATA9 Options + DATA10 Be an option

Step length suggestions 0.9 Master of Science Suggestions on the duration of the
interval 144Master of Science, there is Some needs 216 Master of Science It's normal

		PULSE=0,SQH=0.DATA0-DATA1,SEC0-SEC7 Be invalid SQH=255 Season SQH Remain invalid (specific agreement requirements-H43 Edition)
5	224~239	Encryption first-line, external SEC DATA6: Byte option = 3 Time output 0, otherwise the output current value. DATA9 Options: Refer to the following DATA9/DATA10 List of options DATA10 Options: Refer to the following DATA9/DATA10 List of options Special frame = 224+DATA9 Options + DATA10 Be an option For professionals, all parameters can be modified: Step length suggestions 0.9Master of Science Suggestions on the duration of the interval 144Master of Science, there is Some needs 216Master of Science It's normal PULSE,SQH,DATA0-DATA1,SEC0-SEC7 All can be modified.
6	240~255	Special frame: 241:ATN15 Byte (PULSE,SQH,DATA0-DATA1,SEC0-SEC7 Invalid) 242:30 No. One Line, File in DATA4
		243: First line: 0x52,0x51 244: F2 First line, SEC0=0 Express F0, its his numbers indicate F2 245: Alarm light synchronous output, 246: In use 485 Special frame must be in the interface = 246 Convenient for computers 485 Join 247:15 Byte first line, SQH=255 Season SQH Remain invalid (specific agreement requirements-H43 Edition) 248:13 Byte first line, SQH=255 Season SQH Remain invalid (specific agreement requirements-H43 Edition) 249-NOSQH, 250-YJ First-line, SEC0=255 Allow 24 Foot selection pulse/first-line pass 251-PD0 The test is on the front line. 252-PA15 The first line of detection 253:DY The first line,

254: without Bluetooth 485 All parameters can be modified.

Parameter	Parameter group							
Parameter group code	31	26	5					
Step length	0.9	0.9	0.9					
Interval frame ms	144	216	144					
Special frame	248	247	247					
PULSE	0	0	0					
SQH	0	0	0					
DATA0	(84) 0x54	0	7					
DATA1	(83) 0x53	0	0					
SEC0	0	0	156					
SEC1	0	0	247					
SEC2	0	0	207					
SEC3	0	0	202					
SEC4	0	0	187					
SEC5	0	0	11					
SEC6	0	0	170					
SEC7	0	0	127					
Parking lot File	3	1	1					

		position								
		Side support position	2	1	8					
		Turn the position	0	8	8					
		Anti-theft location	8	0	8					
		Current coefficient	64	64	64					
		Byte option	3	3	3					
		Byte option	0	1	2	3				
		DATA6	0	Electric current	Electric current	Electric current				
		DATA9	0-voltage	Battery	Battery	0				
		DATA10	Battery	Percentage of current	Voltage	0				

DATA9/DATA10 Configuration options table

DATA9 option	DATA9	DATA10 option	DATA10
0	Byte option = 0: voltage unit 1V Byte option = 1:- Byte option = 2: Voltage unit 1V Byte option = 3:-	0	Byte option = 0: power Byte option = 1:- Byte option = 2: the second set of battery power Byte option = 3:-

4	Byte option = 0: power Byte option = 1: power + 0x80 Byte option = 2: the first set of batteries Byte option = 3:-	1	Byte option = 0: current ratio Byte option = 1:- Byte option = 2:- Byte option = 3:-
8	Byte option = 0: Voltage Unit 0.5V Byte option = 1: Voltage Unit 0.5V Byte option = 2: A Group rated voltage Byte option = 3: Centigrade Group rated voltage	2	Byte option = 0: voltage unit 1V Byte option = 1:- Byte option = 2:- Byte option = 3:-
12	(The new version is changed to power +0x80)	3	Byte option = 0: A Group rated voltage Byte option = 1: B Group rated voltage Byte option = 2: Centigrade Group rated voltage Byte option = 3: 0

Special frame = base +DATA9 Options +DATA10 Options.

P position	0-3: Byte 2 Taxi BIT Place 8: It is not displayed. 11: Byte 4 Taxi BIT Place 3 (H40 Increase) 13: Byte 2=0x0a/0x08 (H40 Increase) 14: Byte 2=0x0E (H40 Increase) 15: Bytes 2=1/8 Switch (H40 Increase)	
Side support position	0-3: Byte 2 Taxi BIT Place 7: Byte 3 Taxi BIT7 8: It is not displayed.	
Turn the	0-3: Byte 2 Taxi BIT Place	

position	8: It is not displayed.	
Anti-theft location	0-3: Byte 2 Taxi BIT Place 8: It is not displayed.	

6.2.6 Spa Output signal description:

1	Special frame <16	Output analog voltage
2	Special frame >=16, OBD Effective	Do OBD Alarm indicator
3	Special frame >=16, OBD Invalid, new country The standard speeding prompt is valid.	High-voltage output when speeding
4	Special frame >=16, OBD Invalid, new country The standard speeding prompt is invalid.	When there is an alarm, output the alarm pulse. When there is no alarm, P Output voltage under the gear

6.2.7 DATA0:

HEAD Tacitly approve 8. Adopt other values when customizing

6.2.8 DATA1 :

HEAD2, Default 97, use SEC0~SEC7 = 0

6.2.9 SEC0~SEC7:

Tacitly approve 0, Custom = non 0

6.2.10 P position:

0-3: P File in bytes 2 Taxi BIT Place

7: Parking lot File in bytes 5 Taxi BIT Place 7 (V40 Increase) 8: P The file is not displayed.

11: P File in bytes 4 Block the position 0 (V57 Increase)

13: P File in bytes 2=0x0a/0x08 (V40 Increase) 14: P File in bytes 2=0x0E (V40 Increase)

15: P File in bytes 2=1/8 Switch (V40 Increase)

6.2.11 Side stay position

0-3: Side support in the byte 2 Taxi BIT Place

7: Support in the byte 3 Taxi BIT7

8: The side support is not displayed.

6.2.12 Turn the position

Turn the control display bit in the first-line pass (0~3, default 3), If you don't need to set it to 8

6.2.13 Anti-theft location

Anti-theft indicator display bit in the first line (0~3, default 8), If you don't need to set it to 8

6.2.14 Current coefficient

Tacitly approve 64, 640=0.1A, 320=0.2A 128=0.5A, 64=1A, 32=2A,

6.2.15 Byte option

0,1, 2, 3: Contact the FarDriver adjustment parameters

6.2.16 First-line debugging

Most of the first-line on the market default 21, byte option =3 can basically show the speed, some meters can not show. You can use the first line but do not know the protocol, the step size is recommended 0.9ms, the interval time is recommended 216ms, try the first line to see whether the instrument has feedback:

- 1) Try the special frame first 16, byte option = 3,
- 2) Try the parameter group first. 1: PUSLE=0,SQH=0,DATA0=8,DATA1=97.
- 3) Try the parameter group if it is not a rule 2: PUSLE=0,SQH=0,DATA0=89,DATA1=66.
- 4) If it is not, there are other parameter groups of parameters. See you for details.The table of noodles.
- 5) If all parameter groups are invalid, enter 3.2.
- 6) Special frame 16 If there is no response, change it to 16,32,48,64,80,96,112, 128,144,160, 176,192,208.
- 7) Under normal circumstances, you can find the correct first-line pass by trying the above special frames. The voltage and power are not displayed according to the details. For small-scale modification of special frames can meet the requirements: for example, find 48 The speed can be displayed normally, and the voltage or electricity. The stream display is wrong, so you can try to modify the special frame. 48~63 The number between them is corrected. For example, find it. 160 Fast The degree can be displayed normally, but the voltage or current is not displayed correctly, so you can try it.Try to modify special frames 160~175 Go to The numbers are corrected. Right Parking lot The file display is inaccurate and can be modified. Parking lot Crosspiece location to meet the display requirements.

8) If the above operations cannot be displayed correctly, consider the step length of 0.5ms. Try again.
3.1-3.3 Steps.

9) None of the above operations can be displayed. Consider special frames. 247 Perhaps 248.

10 If the above doesn't work, you need to contact the FarDriver for specific analysis. 224 Perhaps 240~255
Other special frames.

6.3 Tire speed ratio

Note that it should be set correctly so that the speed mileage calculation will be correct.

6.3.1 Tire width

By means of 120/70/R12 For example, tire width = 120

6.3.2 Tire flat rate

By means of 120/70/R12 For example, tire flatness = 70

6.3.3 Hub R

By means of 120/70/R12 For example, hub R = 12

6.3.4 Transmission speed ratio

Hall version hub motor polar logarithm = 20 When, press 4 Calculate the transmission speed ratio to the pole = $20/5 = 5$

6.3.5 Mileage

The total mileage of the controller.

7. Protect

- Protect	
OverVolProtect:90.7V	OverVolRestore:88.7V
LowVolProtect:58V	LowVolRestore:60V
MotorTempProtect:130°C	MotorTempRestore:110°C
ControllerTempProtect:100°C	ControllerTempRestore:80°C
0 BattCoeff:896	Full BattCoeff:1250
ThrottleLost:0-Inval	ThrottleInsert: 256
BackP_Time:18-240s	ReleaseToSeat: 2s
BlockTime: 10s	ParkTime: 10s
LmtSpdStartCap: 0	LmtSpdMinCap: 0
LmtSpdMaxCoeff: 128	TurtleSpeedCurrCoeff: 53
BattSignal:3-Lithium Battery	LowVol Way: 0-Vol2V
IntRes:264	TempCoeff:300

7.1 Voltage protection

7.1.1 Over voltage protection and recovery,

Internally set according to the rated voltage

7.1.2 Under voltage protection, recovery

When the battery voltage is close to the under voltage protection point, the controller reduces the power output, so that the battery will not be damaged by excessive discharge. The general battery under voltage settings are as follows:

Rated voltage	48V	60V	72V	84V	96V	108V
Under voltage protection point	42V	52.5V	63V	73.5V	84V	94.5V

7.1.3 Under-pressure method

2V: Higher than the under voltage point 2V The above power does not decrease, and it reaches the under voltage point +2V. It's time to start to work.Rate.

4V: Higher than the under voltage point. 4V The above does not reduce the power and reaches the under voltage point +4V The power begins to decrease.

8V: Higher than the under voltage point 8V The above does not reduce the power and reaches the under voltage point +8V. The power begins to decrease.

12V: higher than the under voltage point 12V The above does not reduce the power and reaches the under voltage point +12V. The power begins to decrease.

16V: higher than the under voltage point 16V The above does not reduce the power, and it reaches the under voltage point +16V The power begins to decrease.

Turtle speed power 5%: battery capacity is less than or equal to 20% power reduction, less than or equal to 5% use the turtle to go home quickly.

Turtle speed power 6%: battery capacity less than or equal to 30% power reduction, less than or equal to 6% use turtle speed to go home.

Turtle speed power 7%: battery capacity is less than or equal to 40% reduction rate, less than or equal to 7% use the turtle speed to go home.

Turtle speed power 8%: battery capacity is less than or equal to 50% power reduction, less than or equal to 8% use the turtle to go home quickly.

Turtle speed power 9%: battery capacity is less than or equal to 60% power reduction, less than or equal to 9% use the turtle to go home quickly.

Turtle speed power 10%: battery capacity is less than or equal to 70% power reduction, less than or equal to 10% use turtle speed to go home.

SOP Value: According to BMS/CAN Maximum allowable line current received by the bus SOP The value limits the power. Others:

7.2 Temperature protection

7.2.1 Motor temperature protection and recovery:

Internal settings

7.2.2 Controller temperature protection and recovery:

Internal settings

7.2.3 Temperature coefficient

Tacitly approve 300=256+32+12 (H84 Only with new functions)

	Function		
+1	Magnetic brake switch	No addition	Detect the electric door lock voltage
+2/4/6/8/10/12/14	+2 Only 1 Battery = medium speed Battery SOC Large error = low speed Small error allows full speed. +4 Limited to medium speed +6 No restrictions on gears +8 Single battery limit BST +10 Retain +12: Default, reserved	No addition	The battery does not limit the gear.
+16	Dual battery SOP Enable The maximum current is the output value of the battery. Restrictions, note It CAN be used only with the CAN protocol	No addition	The current is not affected by the battery. CAN Issued SOP Impose restrictions on
+32	Tacitly approve	No addition	Special: Follow forward and backward, and With coefficient = energy feedback

			coefficient
+64	Back does not enable follow.	No addition	Back to enable follow
+128	Pinch the brake clutch (Brake sensor required)	No addition	No brake clutch function
+256	Default: Enable acceleration control	No addition	Shielding acceleration control
+512	The controller cooperates with verification, if not verified Certificate, voltage alarm	No addition	The controller does not require verification.
+1024	Low-speed, medium-speed high-speed gear and economic line Sexual sports mode binding	No addition	Unbound, the economy is specified by parameters. Display motion mode
+2048	Low-speed motor shutdown		Do not add by default.
+4096	Switch the brake signal	No addition	Tacitly approve
+8192	Pinch 3 Secondary brake solution Parking lot	No addition	Tacitly approve
+16384	Long press Parking lot Fail to understand Parking lot	No addition	No restrictions
+32768	Long press back and don't need the accelerator.	No addition	You need a gas door to step back.

7.3 Functional protection

7.3.1 Turn to report the loss:

Valid/invalid

7.3.2 Throttle plugging and unplugging Report an emergency:

	Function		
+1	The brakes don't stop when there is a throttle.	No addition	Normal brakes and power failure
+2	Inside the first line D3BIT0 Show push Line mode	No addition	Normal front-line display
+4	The function of the charging foot is changed to the motor direction cut. Change the function	No addition	Maintain the function of charging feet
+8/+16 /+24	Throttle plugging and unplugging 8 (Prevent plugging and unplugging) Caused flying car)	No addition	
+32	Start a single round TCS Function	No addition	Shielding single wheel TCS Function
+64/+128 /+192	Special speed limit function of anti-theft device	No addition	National standard anti-theft function
+256	Default: The throttle is not protected.	No addition	Throttle plug-in protection
+512	The gear is flashing	No addition	Tacitly approve
+1024	Speed gear button delay 1 Second	No addition	Tacitly approve
+2048	The throttle is not protected.	No addition	Tacitly approve
+4096	Changing the gear can be removed. Parking lot Crosspiece	No addition	Tacitly approve

+8192	H938: Side support can be solved Parking lot Crosspiece	No addition	Tacitly approve
+16384	Exclusive to the new control NXT,MZ	No addition	Tacitly approve
+32768	R Version controller single frequency	No addition	R Version controller dual-band

7.3.3 Turn round Parking lot Free time:

Tacitly approve Ten Second

7.3.4 Bucket delay:

Tacitly approve 1 Second

7.3.5 Block time:

Unit 0.1 Second, set 50 Exactly 5 Seconds.

7.3.6 Parking time:

Tacitly approve 0.1 Seconds, the maximum 132 Seconds; Take notice of 0~131 The parking will be cancelled when the time comes., and set 132 Second meeting Parking for a long time will not be cancelled until the controller or motor is overheated.

7.4 Battery protection

7.4.1 Starting power of the speed limit:

The starting power of the power speed limit algorithm

7.4.2 Speed limit power limit:

The minimum power of the power limit algorithm

7.4.3 Speed limit coefficient:

According to the limit coefficient of the power speed limit algorithm

7.4.4 Turtle speed current limiting coefficient:

The turtle speed current limiting coefficient is default. 53, turtle speed current = user's maximum current from the factory * turtle speed current coefficient / 2048;

7.5 Battery characteristics

7.5.1 Battery signal source

Be an option	Content	Explain
0	First-line communication	The SOC of the battery is obtained through the first-line pass
1	Serial port	The SOC of the battery is obtained through the serial port
2	CAN	The SOC of the battery is obtained through the CAN
3	Lithium battery	The characteristics of ternary lithium battery were calculated to simulate SOC
4	Lead acid	SOC was simulated by the calculation of lead-acid battery characteristics
5	Lithium iron (H72)	The SOC was simulated by calculating the characteristics of lithium iron phosphate battery

7.5.2 Under-pressure method

Internal settings

7.5.3 Battery internal resistance

Basic value: 0~255, default 8

	Function		
+256	Measure the second throttle/brake voltage	No addition	Detect the electric door lock voltage
+512	PD Negotiate	No addition	OP Negotiate
+1024	Negotiate 2		
+1536	Negotiate 3		
+2048	Negotiate 4		
+2560	Negotiate 5		
+3072	Negotiate 6		

+3584	Negotiate 7		
+4096	8 Double PID, Ordinary users should not update. Change, otherwise it will easily damage the controller.		Current sensor type >=2 Not supported
+8192	2 Double PID		Current sensor type >=2 Not supported
+12288	4 Double PID		Current sensor type >=2 Not supported

Basic value = 16053/16759 Allow special weak magnetism (deep weak magnetism)

8. PID Control

- PID Paras

AN:10	LM:22en-CN
StartKI:4	StartKP:40
MidKI:8	MidKP:80
MaxKI:12	MaxKP:120
SpeedKI:2	SpeedKP:6
MOE:0-Enable	CurveTime 100ms

8.1 AN:

Motor body characteristics AN Value, parameter range 0~16.

Standard table and motor AN=0. Standard IPM Electric machinery AN=16, QS middle driver suggest set 10. This parameter setting must be in line with the characteristics of the motor. Wheel hub motor, table sticker middle motor, AN Less than 8. Embedded central machine AN The value is not less than 8. Coding with FarDriver The middle motor and the automobile permanent magnet synchronous motor are all adopted. AN=16.

All hub motors on the market are table motors, AN The value is generally set to 0, No more than 4. AN If the value is set incorrectly, the starting efficiency will be lower. Even appear MOE/OVER Protection.

8.2 LM:

The vehicle motor acceleration matching parameters are used to adjust the smooth operation of the motor on the vehicle. The default setting of the car is 22, electric motorcycle default 18. Some low-power tricycles Ten The following is more appropriate.

However, there are individual motor types and the vehicle is poorly matched, starting the low speed segment, the medium speed segment will feel the obvious resonance jitter. Adjusting LM values will improve.

Start from 22, if the low speed section accelerates the jitter, then reduce the LM, start from 16,14,12,11,8,5 to test the effect, those numbers in the middle will also play a role, general consideration would rather be larger, try not to be too small. Too small will not control the current, cause MOE/OVER protection, and even burn control. Therefore, the LM value after the jitter disappears is the best parameter, and it should not be reduced.

Some motors and vehicles are very smooth when LM=22, but after being reduced, it will bring jitter, so pay attention to the fact that there is no problem in the case of LM=22.

Or find that after the jitter resonance appears, the LM value is reduced from 22 to 16,14,... Even 5 does not have much effect, it means that it has nothing to do with this parameter, and it must be changed back to the maximum value, such as 22, rather than a random number retained in the controller.

8.3 PID Parameters: StartKI, MidKI, MaxKI / StartKP, MidKP, MaxKP.

Default parameters StartKI=4, MidKI=8, MaxKI=12 / StartKP=40, MidKP=80, MaxKP=120.

The greater the power of the motor, the higher the voltage, PID The smaller it is. PID Parameters cannot be filled in casually, otherwise it will lead to abnormal work. Even burn control. The following are commonly used PID Set the parameter value. Altogether 9 Set, select one of the parameters to match the motor. The whole vehicle will be modified under the guidance of professionals.

	StartKI	MidKI	MaxKI	StartKP	MidKP	MaxKP	
1	1	1	1	Ten	Ten	Ten	Surfboard by default
2	2	2	3	20	20	30	Ultra-high-power motor
3	3	3	4	30	30	40	
4	4	4	6	40	40	60	High power default
5	4	5	8	40	50	80	

6	6	6	9	60	60	90	Medium power motor
7	6	7	Ten	60	70	100	
8	8	8	12	80	80	120	Small and medium power by default
9	8	9	13	80	90	130	
Ten	8	Ten	15	80	100	150	
11	8	11	16	80	110	160	
12	Ten	12	18	100	120	180	
13	Ten	13	19	100	130	190	
14	Ten	14	21	100	140	210	
15	Ten	15	22	100	150	220	
16	16	16	24	160	160	240	Low-power motor

Take notice of PID Improper parameter setting will cause the system to work abnormally, and even appear. MOE/OVER/PHASE Faults, etc. The difference is too big to cause burn control, and special attention should be paid. Now after the program KI is adjusted, the KP default program will automatically adjust and cannot set the KP value by itself.

Pay attention to some low-power motors. PID The parameters exceed the debugging range of the controller, in this case ask extend we to solve it.

8.4 Velocity SKI,SKP

SKI The smallest 1, the largest 18, the car is heavy KI=18, the car is light KI=2, default KI=9, SKP5~20, default Ten

8.5 MOE:

MOE Tacitly approve On Turn on protection and select Off Turn off the protection and note that it is not allowed to be closed under normal circumstances.

8.6 Curve sampling:

By means of Master of Science Sampling for unit interval, Hxx The acceleration curve of the above version 2560 A point, The following version of the acceleration curve in total Have 510 A point, such as settings 100ms Sampling, then the acceleration curve 510 One point, the total time is 256 Seconds/51 Seconds.

9. Produce:

- Product	
ReCurrRatio: 128	FwReRatio: 128
VolSelectRatio: 106	WeakCurrCoeff: 64
Re Acc: 64s	AlarmDelay: 500
RelayDelay:23671ms	RelayOut: 0
BCEnable: 1	SeatEnable: 1
PEnable: 1	AutoBackPEnable: 0
CruiseEnable: 1	EABSEnable: 1
PushEnable: 1	ForseAntiTheft: 0
OverSpeedAlarm: 0	BrakeStillPark: 0
RememberGear: 1	BackEnable: 1
RelayDelay1S: 0	OSpeedSwitch: 0-Invalid
StartIs: 512	FollowSpeed: 0rpm
Curr-Anti-theft:0-Invalid	Anti-theft Pulse:0-Invalid
Temp 70: 2	Fast RE: 1
InverseTime: 36	SlowDownRpm: 512
SlowDownCoeff: 10	RXD:0-AF
LearnThrottle: 24	LearnVolLow:18432
LearnVolHigh:24320	
DeepWeak: 0-Normal	Protocol485: 0

9.1 Regress current ratio:

Modify the maximum current coefficient of the backward acceleration, and the maximum backward current = the maximum bus current * coefficient/128.

9.2 Back teeth ratio:

A parameter required for the forward and reverse gear transmission system. 0~255

	Additional functions		Functions not added
+256	Side support signal reverse	No addition	Side support signal reverse
+512	Allow small current switching	No addition	Small current switching is not allowed.
+1024	Anti-slip in the air Parking	No addition	It's not in the air.
+2048	Record mileage	No addition	Do not record mileage
+4096	Extension code Y Lower	No addition	Extension code Y Down the awesome

	magnetic brake switch		function
+8192	Downhill reverse charging, over voltage and power failure	No addition	Downhill reverse charging control over voltage
+16384	Shielding gear speed limit 2	No addition	Turn on the gear speed limit 2
+32768	PD1 Input	No addition	PD1 Send out

9.3 Voltage selection coefficient:

Select the working voltage of the controller through the pin, and the calculation coefficient when selecting low voltage.

Take for example 72V In the system, after the voltage selection pin is lowered, it becomes 60V System, its coefficient = $60/72 \times 128 = 107$

9.4 Weak magnetic current coefficient

Scope 48~80, default 64

9.5 Regress acceleration coefficient:

Back to the maximum acceleration coefficient by default 32, the largest 224

9.6 Alarm delay:

The recovery delay time setting when the anti-theft alarm generates the alarm, by default 500ms Choose 100 Multiple of.

Use of single digits: 1~9 Express 100RPM~900RPM Embellish one's speech with literary quotationsThe brakes in the speed do not respond. 0 It means that the brakes are responsive.

Ten digits (H59)	First-line display	READY When the light turns	First-line electronic brake state (H60)
0	Brake status	Go out	Show
1	Brake status	Go out	Do not show
2	Brake status	Bright	Show
3	Brake status	Bright	Do not show

4	Brake failure	Go out	Show
5	Brake failure	Go out	Do not show
6	Brake failure	Bright	Show
7	Brake failure	Bright	Do not show
8	Brake status	Go out	Show
9	Brake status	Go out	Do not show

9. 7 Relay delay:

Set the delay closing time of the relay after the system is powered on: ms As a unit, by default 1ms Set it to when needed Ten00ms Wait. Take notice of H44 The new version of the controller is all changed to functional positions:

Place	Project	Function description
BIT0	Enable while supporting	0: Disable the side support function, 1: Enable the side support function, default 1
BIT1	The bucket is enabled.	0: Disable the seat cushion function, 1: Enable the seat cushion function, default 1
BIT2	Parking lot File enabled	0: Disable Parking lot File function, 1: Enable Parking lot File function, default 1
BIT3	Automatic return Parking lot	0: Disable automatic return Parking lot Function, 1: Enable automatic return to Parking lot Function, by default 0
BIT4	Cruise enabled	0: Disable the cruise function, 1: Enable the cruise function, default 1
BIT5	EABS Start using	0: Disable EABS Function, 1: Enable EABS Function, default 1
BIT6	Implementation and activation	0: Disable the assistance implementation function, 1: Enable the assistance implementation function, default 1, Qi When it is implemented, it will be automatically cancelled. Parking lot File entry

		implementation method
BIT7	Forced anti-theft	0: No forced anti-theft, 1: forced entry anti-theft, default 0
BIT8	Speeding alarm	0: Disable the speeding alarm function, 1: Enable the speeding alarm function, default 0
BIT9	The brakes can't stop the car.	0: The brake will remove the parking, 1: The brake will not remove the parking, the default 0
BIT10	Block memory	0: Gear memory is closed 1: Gear memory is effective Tacitly approve 1
BIT11		Tacitly approve 1
BIT12		Tacitly approve 1
BIT13		Tacitly approve 0
BIT14	Back to enable	0: Disable the backward function, 1: Enable the backward function, default 1
BIT15	Relay delay	0: No relay delay, 1: delay 1 Seconds, default 0

Relay delay = $BIT15 * 2^{15} + BIT14 * 2^{14} + \dots + BIT1 * 2^1 + BIT0$;

9. 8 Slow startls:

Default 512

H88 Versions above slow start range: 8~2048. If the motor has a gear reduction mechanism, if the gear meshing gap is large, Adjusting this parameter can improve the phenomenon of teething: number = $N * 32 + m$, $m=0$ It means that there is no gap, $m=1 \sim 15$, between small gap, $M=15$ The gap is the largest, and you don't hit the teeth when the twisting handle is delayed. Medium size Reduce, Turn the teeth when the delay is in force. Medium size Increase; $N=0$ Indicates that the shielding starts slowly, $n=1 \sim 31$ Adjust the fit strength of the teeth and twist it. Don't hit your teeth for a moment. N Increase, twist Punch the teeth in the moment. N Decrease.

9.9 Follow the speed:

Tacitly approve 0, some customers need to 100 Exit and follow within the turn, and set it up. 100

9.10 Current anti-theft,

Tacitly approve 0. It is difficult to push with greater resistance when anti-theft, and does not consume the battery. For 1 At that time, the anti-theft lock motor will lock the electricity. The machine cannot rotate and consumes the battery.

9.11 Anti-theft pulse:

Tacitly approve 0	The anti-theft signal is a level signal, pull down the anti-theft.
1 (speeding alarm = 0)	The anti-theft signal is 1Master of Science Pulse signal, anti-theft if there is a pulse
1 (speeding alarm = 1)	The pulse is the national standard data, and the low level is anti-theft.
2	Power-on speed limit, pulse signal 8 Lift the speed limit

Temperature 70:

Controller temperature protection 70 Algorithm:

0 Express Start control at 50°, maximum time to protect the battery, controller and motor.And extend the driving mileage.

1 Express Start control at 70°, and the temperature of the battery, controller and motor is considered in the mileage and performance.

3 Express Start control at 80°, performance is preferred, but the mileage is the shortest, and rapid acceleration is most likely to cause over temperature protection.

9.12 Reverse time:

Start the slow-down and Parking Time, 3-48, default 36.

Odd number: the gear gap is too large,Parking Flexibility. H78 Even number: the gear gap is small,Parking Normalize H78The reverse time of Hall version 7-11 The effect is already very good, and the minimum is 3. Be careful that if it is too small, it will be easy to over rush. Reverse time of the encoder version 33 The effect is very good, the smallest is 11. Be careful that it is too small to over rush.

	Additional functions		Functions not added
+64	JL National standard status SEC5=48V The maximum speed ratio SEC6=60V The maximum speed ratio SEC7=72V The maximum	No addition	General status

	speed ratio		
+128	Push down 2 Solution Parking lot	No addition	Push down 1 Solution Parking lot
+256	Low and medium speed to high speed	No addition	Default low and medium speed
+512	No sound prompt	No addition	There is a sound prompt.
+1024	Retain	No addition	
+2048	Shield the sound of the buzzer	No addition	The buzzer is enabled
+4096	Turn on the backbuzzer prompt tone	No addition	Block the backbuzzer prompt tone
+8192	Prohibit motor angle sensor repair	No addition	Allow the motor angle sensor to be repaired
+16384	Allow brake fault repair	No addition	Prohibit brake fault repair
+32768	Reverse first-line	No addition	Positive front-line communication

9.13 Slow down the speed:

The critical value of slow speed reduction of steep slope, default 320, can be set 256~1024

This parameter is also used for the adjustment of the implementation speed.

9.14 Slow-down coefficient:

Tacitly approve 2. Can be set 1-7, the larger the number, the slower the slow-down speed. Note that the number is too large, and the gear gap will shake. The smaller the number, the faster the slowdown.

H937 Version anti-slip: slow-down coefficient 2~15, when parking before and after parking Until the festival does not shake before and after.

9.15 0 Quick switch:

Speed is equal to 0 Only when the gear is allowed to switch.

9.16 Deep weak magnetism:

Email: info@siaecosys.com Website: www.siaecosys.com
Address: 188 Kaifa Road east, Development Zone, Taizhou, Zhejiang, China

Battery internal resistance = special value (please consult the FarDriver) In this case, the depth is weak.(At present, the absolute value encoder of a motor factory Special use for motors) It can be set to 1=Valid. Otherwise, it will automatically return. 0=Invalid.

9.17 RS485 Agreement:

Extension code = "X": RS485 Protocol control in the state, connect by default RS485 Instrument, after connecting to the computer, It will automatically enter the communication mode with the upper computer.

0: For the whole vehicle factory YJ Taxi BMS Protocol control

1: For the vehicle factory PD,OP Taxi VCU Negotiate

2: For the whole vehicle factory DP2 No. VCU Protocol control

3: For the whole vehicle factory Xxx VCU Protocol control 4: For the whole car factory ...

5: For the whole car factory TUYA Control

9.18 Self-learning accelerator:

Tacitly approve 24, the largest 36. Pay attention to the requirement that the motor is in a no-load state when learning. Normal motor should be started for self-learning. 24 The throttle is enough. A small number of motors can't turn up. Increase the throttle. 36 To start the motor. If the self-study motor does not turn Move, it is likely that the Hall line is wrong or the phase line is not connected in order. Check the Hall wiring, or put the blue-green phase line.Swap

Try to learn by yourself again.

Self-learning throttle = 23, 24, 25, 26, 27 Corresponds to different idle forces. 23 The weakest, 27 The strongest.

9.19 Low self-learning voltage (self-learning) VQL):

Basic default value 18432, basic wide range value = 25856, normal situation Keep the default value.

Add value	Function		
+1	Start the idle clutch function (H67)	No addition	Do not start the clutch function
+2	Voltage foot =Personal identification number8 Season: (H72) PIN14 Foot Come into contact with Earth Soft Rise Move, PIN8 Come into contact with Earth 60V, suspended 48/72V Automatic	No addition	Voltage foot = PIN8 Time: PIN8,PIN14 Two-wire selection of electricity Press

	voltage switching		
+4	Reverse neutral is the ignition switch (H72)	No addition	Normal reverse neutral function
+8	Knob to adjust the maximum bus bar (H72)	No addition	Normal mode
+16	One-click repair foot to switch half of the current (H72)	No addition	No switching
+32	Pinch 2 Secondary brake solution Parking lot (H72)	No addition	Normal Parking lot
+64	Press P first, then pinch the brake to remove P (H72)	No addition	Normal Parking lot
+128	Single battery is not allowed. BOOST(H74)	No addition	Normal BOOST

9.20 High self-learning voltage (self-learning) VQH):

Basic default value 24320, The basic wide range value is = 31744, and the default value is maintained under normal circumstances.

Add value	Function		
+1	Click to move forward and backward (H67)	No addition	Switch forward and backward
+2	Double throttle voltage (H72)	No addition	Single throttle
+4	Intermediate throttle neutral (H74)	No addition	Normal throttle
+8	One-click repair is effective (H72)	No addition	One-click repair does not work.
+16	Strengthen input signal filtering (H72)	No addition	Normal input filter

+32	Cruise speed is not limited, and acceleration is implemented. Degree controllable (H72)	No addition	Cruise is limited to 3/4 Maximum speed, implementation The slowest
+64	Six times speed Hall detection (H72)	No addition	Single-speed Hall detection
+128	New MTPA Vector (H73)	No addition	Common practice MTPA Vector

10.Calibration(The coefficient of voltage and current of the factory calibration controller cannot be modified)

- FixedParas

LineCurrCoeff:439 BattVoltage:256
 PhaseCoeffA:273 PhaseCoeffB:273
 PhseAZero:2059 PhaseCZero:2058
 TempCoeff:300 VoltageCoeff:256
 Save Times:63

10.1 Line current coefficient:

Default settings

10.2 Bus voltage:

Default settings

10.3 Phase current coefficient A:

Default settings

10.4 Phase current coefficient B:

Default settings

10.5 phase current zero point A:

Default settings

10.6 Zero point of phase current centigrade:

Default settings

10.7 Temperature coefficient:

Default settings

10.8 Voltage coefficient:

Default settings

10.9 calibration coefficient:

Default settings

11. Central control (not enabled)

